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Database Management Systems

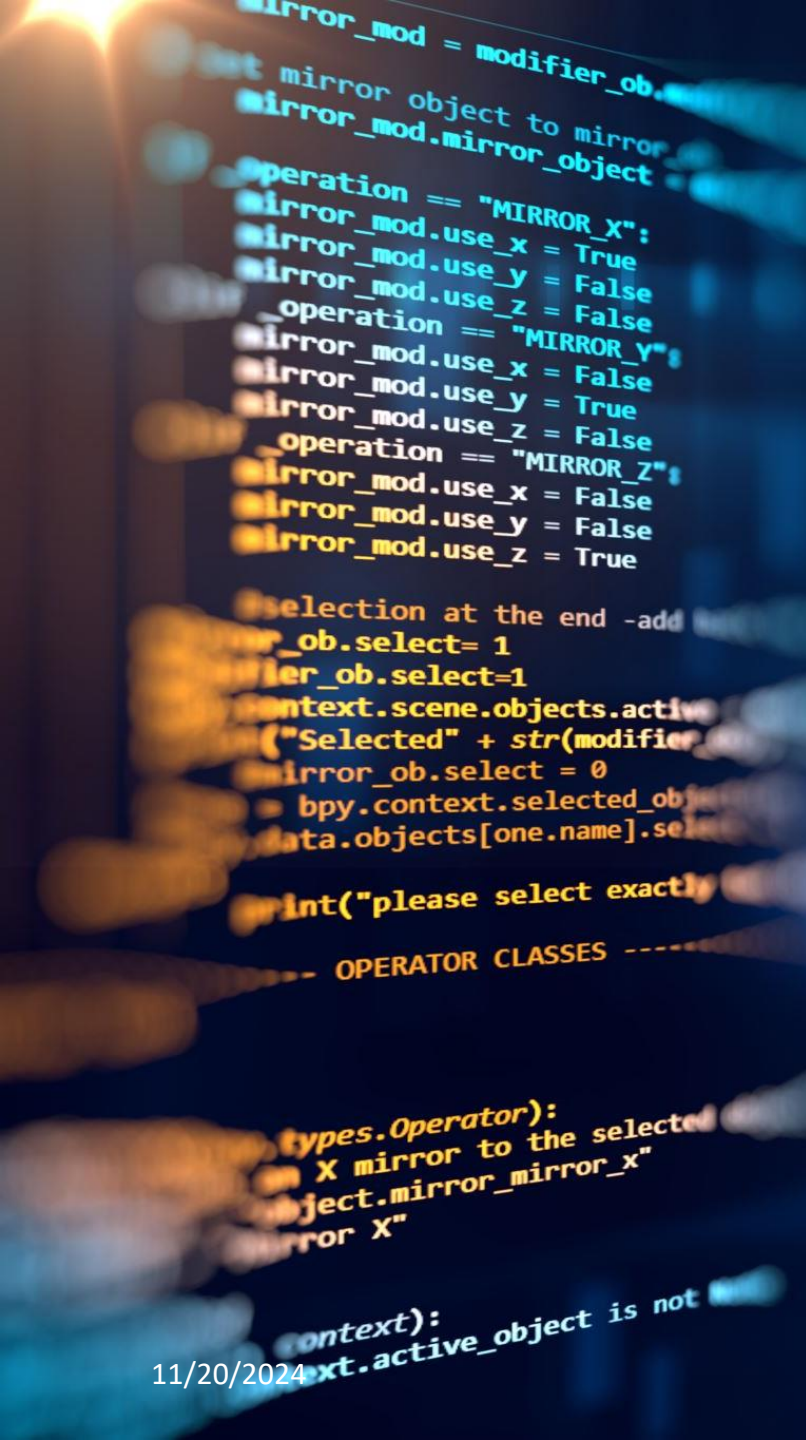
Week 1, Day 3

MySQL—A RDBMS

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Introduction

- SQL is the standard language for dealing with **Relational Databases**.
- SQL is used to insert, search, update, and delete database records.

The following SQL statement selects all the records in the "Customers" table:

Your first Query 🤗

```
1 • SELECT * FROM CUSTOMER
```

customer_id	store_id	first_name	last_name	email
1	1	MARY	SMITH	MARY.SMITH@sa...
1	1	PATRICIA	JOHNSON	PATRICIA.JOHN...
1	1	LINDA	WILLIAMS	LINDA.WILLIAMS...
2	2	BARBARA	JONES	BARBARA.JONES...

Keep in Mind That



- SQL keywords are NOT case sensitive: **select** is the same as **SELECT**

```
SELECT column1, column2, ...  
FROM table_name;
```

The MySQL SELECT Statement

- The SELECT statement is used to select data from a database.
- The data returned is stored in a result table, called the result-set.

SLELECT EXAMPLE

1 • **SELECT** CustomerName, City, Country **FROM** Customers;

Result Grid  Filter Rows: Export:  Wrap Cell Content:  Fetch rows: 			
	CustomerName	City	Country
▶	Alfreds Futterkiste	Berlin	Germany
	Ana Trujillo Empa...	México D.F.	Mexico
	Antonio Moreno ...	México D.F.	Mexico
	Around the Horn	London	UK
	Berglunds snabbk...	Luleå	Sweden

SELECT DISTINCT Statement

- The **SELECT DISTINCT** statement is used to return only distinct (different) values.

SELECT DISTINCT Statment

1 • **SELECT** Country **FROM** Customers;

1 • **SELECT DISTINCT** Country **FROM** Customers;

Country
Germany
Mexico
Mexico
UK
Sweden

Country
Germany
Mexico
UK
Sweden
France

How many countries have our customers?

```
1  SELECT COUNT(DISTINCT Country) as 'Countries' FROM Customers;
```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content: 

Countries
21

WHERE Clause

- The **WHERE** clause is used to filter records.
- It is used to extract only those records that fulfil a specified condition.

```
SELECT column1, column2, ...  
FROM table_name  
WHERE condition;
```

WHERE CLAUSE

```
1 • SELECT * FROM Customers
2 WHERE Country = 'Mexico';
```

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:					
	CustomerID	CustomerName	ContactName	Address	City
▶	2	Ana Trujillo Empa...	Ana Trujillo	Avda. de la Const...	México D.F.
	3	Antonio Moreno ...	Antonio Moreno	Mataderos 2312	México D.F.
	13	Centro comercial ...	Francisco Chang	Sierras de Grana...	México D.F.
	58	Pericles Comidas ...	Guillermo Fernán...	Calle Dr. Jorge C...	México D.F.
	80	Tortuga Restaura...	Miguel Angel Paol...	Avda. Azteca 123	México D.F.

Operators in The **WHERE** Clause

- EQUAL (=)
- GREATER (>)
- LESS (<)
- NOT EQUAL TO (<>)
- BETWEEN (RANGE)
- LIKE (Search pattern)
- IN (TO SPECIFY multiple values)

EQUAL

- 1 • **SELECT** * **FROM** Products
- 2 **WHERE** Price = 18;

ProductID	ProductName	SupplierID	CategoryID	Unit	Price
1	Chais	1	1	10 boxes x 20 bags	18
35	Steeleye Stout	16	1	24 - 12 oz bottles	18
39	Chartreuse verte	18	1	750 cc per bottle	18
40	Boston Crab Meat	19	8	24 - 4 oz tins	18
76	Lakkalikööri	23	1	500 ml	18

GREATER THAN (>)

```
1 • SELECT * FROM Products
2 WHERE Price > 30;
```

ProductID	ProductName	SupplierID	CategoryID	Unit	Price
8	Northwoods Cran...	3	2	12 - 12 oz jars	40
9	Mishi Kobe Niku	4	6	18 - 500 g pkgs.	97
10	Ikura	4	8	12 - 200 ml jars	31
12	Queso Manchego...	5	4	10 - 500 g pkgs.	38
17	Alice Mutton	7	6	20 - 1 kg tins	39

LESS THAN (<)

```
1 • SELECT * FROM Products
2 WHERE Price < 30;
```

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content: Fetch rows:						
	ProductID	ProductName	SupplierID	CategoryID	Unit	Price
▶	1	Chais	1	1	10 boxes x 20 bags	18
	2	Chang	1	1	24 - 12 oz bottles	19
	3	Aniseed Syrup	1	2	12 - 550 ml bottles	10
	4	Chef Anton's Caj...	2	2	48 - 6 oz jars	22
	5	Chef Anton's Gu...	2	2	36 boxes	21

Products 10 x

NOT EQUAL TO (<>)

```
1 • SELECT * FROM Products
2 WHERE Price <> 18;
3
```

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content: Fetch rows:						
	ProductID	ProductName	SupplierID	CategoryID	Unit	Price
▶	2	Chang	1	1	24 - 12 oz bottles	19
	3	Aniseed Syrup	1	2	12 - 550 ml bottles	10
	4	Chef Anton's Caj...	2	2	48 - 6 oz jars	22
	5	Chef Anton's Gu...	2	2	36 boxes	21
	6	Grandma's Boyse...	3	2	12 - 8 oz jars	25

BETWEEN

```
1 • SELECT * FROM Products
2 WHERE Price BETWEEN 50 AND 60;
3
```

t Grid  Filter Rows: Edit:    Export/Import:   Wrap Cell Content: 					
ProductID	ProductName	SupplierID	CategoryID	Unit	Price
51	Manjimup Dried ...	24	7	50 - 300 g pkgs.	53
59	Raclette Courdav...	28	4	5 kg pkg.	55
IL	NULL	NULL	NULL	NULL	NULL

IN

```
1 • SELECT * FROM Customers
2 WHERE City IN ('Paris', 'London');
3
4
```

t Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:					
CustomerID	CustomerName	ContactName	Address	City	Pa
4	Around the Horn	Thomas Hardy	120 Hanover Sq.	London	W
11	B's Beverages	Victoria Ashworth	Fauntleroy Circus	London	EC
16	Consolidated Hol...	Elizabeth Brown	Berkeley Gardens...	London	W
19	Eastern Connection	Ann Devon	35 King George	London	W
53	North/South	Simon Crowther	South House 300...	London	SV

AND, OR and NOT Operators

- The WHERE clause can be combined with AND, OR, and NOT operators.
- The AND and OR operators are used to filter records based on more than one condition:
- The AND operator displays a record if all the conditions separated by AND are TRUE.
- The OR operator displays a record if any of the conditions separated by OR is TRUE.
- The NOT operator displays a record if the condition(s) is NOT TRUE.

AND

```
1 • SELECT * FROM Customers
2 WHERE Country = 'Germany' AND City = 'Berlin';
3
4
```

Result Grid  Filter Rows: Edit:    Export/Import:   Wrap Cell Content: 						
CustomerID	CustomerName	ContactName	Address	City	Po	
4	Around the Horn	Thomas Hardy	120 Hanover Sq.	London	W	
11	B's Beverages	Victoria Ashworth	Fauntleroy Circus	London	EC	
16	Consolidated Hol...	Elizabeth Brown	Berkeley Gardens...	London	W	
19	Eastern Connection	Ann Devon	35 King George	London	W	
53	North/South	Simon Crowther	South House 300...	London	SW	

OR

```
1 • SELECT * FROM Customers
2 WHERE City = 'Berlin' OR City = 'Stuttgart';
3
4
```

esult Grid   Filter Rows: Edit:    Export/Import:   Wrap Cell Content: 					
CustomerID	CustomerName	ContactName	Address	City	PostalCode
1	Alfreds Futterkiste	Maria Anders	Obere Str. 57	Berlin	12209
86	Die Wandernde K...	Rita Müller	Adenauerallee 900	Stuttgart	70563
NULL	NULL	NULL	NULL	NULL	NULL

NOT

```
1 • SELECT * FROM Customers
2 WHERE NOT Country = 'Germany';
3
4
```

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content: Fetch rows:				
CustomerID	CustomerName	ContactName	Address	City
2	Ana Trujillo Empa...	Ana Trujillo	Avda. de la Const...	México D.F.
3	Antonio Moreno ...	Antonio Moreno	Mataderos 2312	México D.F.
4	Around the Horn	Thomas Hardy	120 Hanover Sq.	London
5	Berglunds snabbk...	Christina Berglund	Berguvsvägen 8	Luleå
7	Blondel père et fils	Frédérique Citeaux	24, place Kléber	Strasbourg

Question
